Indian Institute of Technology Gandhinagar

Department of Computer Science and Engineering

Project Course Report

Indian Institute of Technology Gandhinagar

Palaj, Gandhinagar – 382355

Fairness in Federated Learning: A Comprehensive Study on Dynamic Client-Adaptive Fairness Weighting

Submitted by

Yash Goyal (24110399)

Email: yash.g@iitgn.ac.in

Under the guidance of

Prof. Manisha Padala

Professor, Department of Computer Science and Engineering

Indian Institute of Technology Gandhinagar

Email: manisha.padala@iitgn.ac.in

April 12, 2026

Abstract

This comprehensive project investigates the intersection of Federated Learning and machine learning fairness, specifically focusing on gender classification with attention to demographic disparities across racial groups. Conducted over four months under the direction of Prof. Manisha Padala, this study progresses from foundational Federated Learning concepts through the identification of inherent fairness issues in decentralized learning systems to the development and validation of a novel solution: Dynamic Client-Adaptive Fairness Weighting (DCA-FW). The research demonstrates that standard Federated Learning exhibits significant race-wise accuracy disparities, which are further amplified by adversarial attacks. The proposed DCA-FW method successfully reduces the fairness gap (from 0.0852 to 0.0507 on UTKFace and from 0.0992 to 0.0812 on FairFace) while maintaining stable overall accuracy, proving that client-level adaptive mechanisms are highly effective for addressing demographic bias in decentralized learning environments.

Contents

1 Introduction **4**

1.1 Background and Motivation 4

1.2 Project Objective 4

1.3 Project Scope and Timeline 5

2 Theoretical Foundations and Background **6**

2.1 Federated Learning: Core Concepts and Definitions 6

2.1.1 Federated Learning Architecture 6

2.1.2 Data Distribution Concepts: IID vs Non-IID 7

2.1.3 Vulnerability to Adversarial Attacks in FL 8

2.2 Fairness in Machine Learning: Definitions and Metrics 8

2.2.1 Fairness: Core Concepts 8

2.2.2 Fairness Metrics Used in This Study 8

2.2.3 Sources of Bias in Machine Learning 9

3 Dataset Descriptions and Experimental Setup **10**

3.1 Dataset Overview 10

3.1.1 MNIST Dataset (Foundational Phase) 10

3.1.2 UTKFace Dataset (Primary Research Phase) 10

3.1.3 FairFace Dataset (Validation Phase) 11

3.2 Experimental Setup and Hyperparameters 11

3.2.1 Model Architecture 11

3.2.2 Federated Learning Configuration 11

4 Phase 1: Foundational Federated Learning Understanding **13**

4.1 Objectives and Motivation 13

4.2 Experimental Design 13

4.2.1 Experiment 1: Baseline FL with IID Data Distribution . 13

4.2.2 Experiment 2: Non-IID Data Distribution Impact 14

4.2.3 Experiment 3: Label-Flipping Attack on Non-IID Data . 14

4.3 Transition to Fairness-Focused Research 15

5 Phase 2: Fairness Issues Identification 16

5.1 Objectives 16
5.2 Baseline FL Results (Clean Setup) 16
5.2.1 Overall Performance 16
5.2.2 Race-Wise Accuracy Analysis 16
5.3 Adversarial Attack Scenario 17
5.3.1 Attack Configuration 17
5.3.2 Results Under Attack 17

6 Phase 3: Fairness Improvement Methods 18

6.1 Overview of Fairness Intervention Strategies 18
6.2 Method 1: Static Weighted Cross-Entropy 18
6.2.1 Concept and Implementation 18
6.2.2 Results 19
6.2.3 Analysis of Failure 19
6.3 Method 2: Fairness Regularization 19
6.3.1 Concept and Implementation 19
6.3.2 Results 19
6.3.3 Analysis of Failure 20
6.4 Method 3: Adversarial Fairness Learning 20
6.4.1 Concept and Implementation 20
6.4.2 Results 20
6.4.3 Analysis of Performance 20
6.5 Method 4: Dynamic Client-Adaptive Fairness Weighting (DCA-FW) 21
6.5.1 Concept and Mechanism 21
6.5.2 Key Innovations 21
6.5.3 Results 22
6.5.4 Why DCA-FW Succeeds 22
6.5.5 Fairness Client Selection 22
6.5.6 Weight Computation Details 23
6.5.7 Global Model Aggregation 23
6.6 Method 5: Attack-Based Fairness Approach 23
6.6.1 Concept and Motivation 23
6.6.2 Results 23
6.6.3 Analysis of Failure 23
6.7 Comparative Summary of All Methods 23

7 Phase 4: Validation on FairFace Dataset **25**
7.1 Motivation for Multi-Dataset Validation 25
7.2 FairFace Baseline FL Results 25
7.3 FairFace with DCA-FW - Multiple Runs 26
7.3.1 Run 1: Fairness Clients {0, 1, 4, 9} (40%) 26
7.3.2 Run 2: Fairness Clients {0, 1, 4, 6, 9} (50%) 26
7.3.3 Run 3: Fairness Clients {0, 1, 4, 5, 6, 9} (60%) 26
7.4 Analysis of Multi-Run Consistency 27

8 Conclusions and Future Work **28**
8.1 Key Findings 28
8.1.1 Federated Learning Inherently Exhibits Fairness Disparities 28
8.1.2 Traditional Fairness Methods Fail in FL Settings 28
8.1.3 Client-Level Adaptive Mechanisms are Essential 28
8.1.4 Fairness Improvement Without Accuracy Sacrifice 29
8.1.5 Method Generalizes Across Datasets 29
8.2 Implications for Federated Learning Systems 29
8.2.1 Design Implications 29
8.2.2 Practical Implementation Considerations 29
8.3 Limitations 29
8.4 Future Research Directions 30
8.4.1 Technical Extensions 30
8.4.2 Application Domains 30
8.5 Final Summary 30

A DCA-FW Algorithm Pseudocode **31**

B Results Summary Tables **32**
B.1 Complete Results Comparison 32

Chapter 1

Introduction

1.1 Background and Motivation

Machine learning has become deeply integrated into critical decision-making systems across numerous domains, from healthcare diagnostics to credit scoring and criminal justice. However, these systems frequently exhibit systematic biases that disproportionately affect certain demographic groups, leading to unfair treatment and perpetuating societal inequalities. As machine learning models become more distributed and decentralized, particularly through Federated Learning paradigms, understanding and mitigating fairness issues becomes increasingly critical.

Federated Learning (FL) has emerged as a privacy-preserving alternative to centralized machine learning, allowing multiple clients (devices, organizations, or individuals) to collaboratively train a global model without sharing their raw data. While FL addresses privacy concerns inherent in traditional centralized learning, it introduces new challenges for ensuring fairness. The decentralized nature of FL, combined with non-identical and non-independent (Non-IID) data distributions across clients, can amplify demographic biases and create disparities in model performance across demographic groups.

1.2 Project Objective

The primary objective of this research is to improve fairness in gender classification using Federated Learning while maintaining or improving overall model accuracy. Specifically, the project aims to:

1. Reduce race-wise accuracy disparity (measured as the fairness gap between the highest and lowest performing racial groups)

2. Maintain or improve overall model performance across all demographic groups
3. Develop a scalable and adaptive fairness method suitable for federated learning environments
4. Validate the proposed method across multiple datasets to demonstrate generalization

1.3 Project Scope and Timeline

This four-month research initiative follows a structured, iterative approach with weekly milestone reviews. The project is organized into four distinct phases:

- **Phase 1:** Foundational FL Understanding using MNIST (Month 1)
- **Phase 2:** Fairness Issues Identification on UTKFace (Month 1-2)
- **Phase 3:** Fairness Improvement Methods Development (Month 2-3)
- **Phase 4:** Validation on FairFace Dataset (Month 3-4)

Chapter 2

Theoretical Foundations and Background

2.1 Federated Learning: Core Concepts and Definitions

2.1.1 Federated Learning Architecture

Federated Learning is a decentralized machine learning paradigm where a global model is trained collaboratively across multiple independent clients without centralizing data. The fundamental architecture consists of:

Clients: Independent nodes (devices, organizations) that hold local data and perform local computation. Each client maintains its own dataset and trains a model locally.

Server (Central Aggregator): Coordinates the training process by aggregating model updates from clients and broadcasting the updated global model back to clients.

Communication Rounds: Iterative cycles where clients train locally and communicate updates to the server.

The Federated Averaging Algorithm (FedAvg) is the standard aggregation mechanism, mathematically defined as:

$$\mathbf{w}^{(t+1)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathbf{w}_k^{(t+1)} \tag{2.1}$$

where:

- $\mathbf{w}^{(t+1)}$ is the updated global model at round $t+1$
- n_k is the number of samples at client k
- n is the total number of samples across all clients
- $\mathbf{w}_k^{(t+1)}$ is the updated local model at client k after training for one round

Key advantages of Federated Learning include:

- Privacy preservation: Raw data never leaves the client
- Communication efficiency: Only model parameters are transmitted
- Scalability: Can accommodate thousands of clients
- Compliance: Meets regulatory requirements for data protection

2.1.2 Data Distribution Concepts: IID vs Non-IID

An essential concept in understanding Federated Learning behavior is the distribution of data across clients.

IID (Independent and Identically Distributed) Data

IID data assumes that samples are randomly distributed across clients, such that each client has a representative sample of the overall data distribution. In IID settings, the distribution $P(X, Y)$ is similar across all clients. This is the idealized scenario where clients have similar data distributions and model convergence is fastest.

Non-IID (Non-Independent and Non-Identically Distributed) Data

Non-IID data represents the more realistic scenario where different clients have different data distributions due to natural geographical, demographic, or behavioral variations. In Non-IID settings:

- Different clients may have different class distributions
- Certain clients may have only a subset of classes
- Feature distributions may differ across clients
- Data is imbalanced differently on each client

Non-IID data significantly complicates Federated Learning:

- Slower convergence to the global model
- Potential accuracy degradation
- Increased vulnerability to attacks
- Exacerbation of demographic biases

2.1.3 Vulnerability to Adversarial Attacks in FL

Federated Learning systems are inherently vulnerable to poisoning attacks, where malicious clients introduce corrupted updates to degrade model performance. Label-flipping attacks represent one straightforward attack vector, where a malicious client deliberately flips labels from one class to another.

The impact of such attacks in Non-IID settings is particularly severe: if a malicious client is the only source of data for a particular class and it flips those labels, the global model may learn the incorrect mapping entirely, collapsing performance for that class to near zero.

2.2 Fairness in Machine Learning: Definitions and Metrics

2.2.1 Fairness: Core Concepts

Fairness in machine learning refers to the absence of unfair bias or discrimination against individuals or groups based on protected attributes such as race, gender, age, or other demographic characteristics. Machine learning model fairness can be viewed through multiple lenses:

Demographic Parity: The model makes similar predictions for different demographic groups, regardless of ground truth.

Equality of Opportunity (Equalized Odds): The model achieves similar true positive rates and false positive rates across different demographic groups.

Individual Fairness: Similar individuals (according to some criterion) receive similar treatment from the model.

2.2.2 Fairness Metrics Used in This Study

Statistical Parity Difference (SPD)

SPD measures whether a model's positive prediction rate is similar across demographic groups. Formally:

$$\text{SPD} = \left| P(\hat{Y} = 1|\text{Group}_A) - P(\hat{Y} = 1|\text{Group}_B) \right| \tag{2.2}$$

where $\hat{Y}$ is the model's prediction. SPD ranges from 0 (perfectly fair) to 1 (maximally unfair). A value close to 0 indicates the model treats different demographic groups similarly in terms of positive predictions.

Equalized Odds Difference (EOD)

EOD measures the difference in true positive rates (also called equality of opportunity) and false positive rates across demographic groups. Formally:

$$\text{EOD} = |\text{TPR}_A - \text{TPR}_B| + |\text{FPR}_A - \text{FPR}_B| \tag{2.3}$$

where $\text{TPR} = \frac{\text{TP}}{\text{TP+FN}}$ represents the proportion of positive cases correctly identified, and $\text{FPR} = \frac{\text{FP}}{\text{FP+TN}}$ represents the proportion of negative cases incorrectly identified as positive.

A lower EOD value indicates fairer performance across demographic groups.

Fairness Gap (Maximum Accuracy Disparity)

The fairness gap is defined as the difference between the highest and lowest group-wise accuracy:

$$\text{Fairness Gap} = \max_i(\text{Accuracy}_{\text{race}_i}) - \min_i(\text{Accuracy}_{\text{race}_i}) \tag{2.4}$$

This metric is intuitive and directly interpretable: it measures the maximum disparity in model performance across demographic groups.

2.2.3 Sources of Bias in Machine Learning

Technical bias can arise from multiple sources:

- **Data Bias:** The training dataset may not be representative of all demographic groups, or may contain imbalanced representation.
- **Algorithmic Bias:** The model architecture or training procedure may inherently favor certain groups over others.
- **Feature Bias:** The features used by the model may correlate with protected attributes in ways that introduce unfair discrimination.
- **Non-IID Data Distribution:** In Federated Learning, different clients may have fundamentally different class and feature distributions.

Chapter 3

Dataset Descriptions and Experimental Setup

3.1 Dataset Overview

3.1.1 MNIST Dataset (Foundational Phase)

Dataset Specifications:

- Total Images: 70,000 (60,000 training, 10,000 test)
- Image Dimensions: 28×28 grayscale
- Classes: 10 digit classes (0-9)
- Nature: Handwritten digits dataset

MNIST was selected for the foundational phase because it provides a controlled environment for understanding FL concepts without the complexity of real-world fairness issues.

3.1.2 UTKFace Dataset (Primary Research Phase)

Dataset Specifications:

- Total Images: $\approx 23,695$
- Train/Test Split: 80/20
- Demographic Attributes: Gender (Male/Female), Race (5 categories)
- Task: Binary gender classification with race-based fairness analysis

- Race Distribution:
  - White: 10,069 (42.49%)
  - Black: 4,526 (19.10%)
  - Asian: 3,433 (14.49%)
  - Indian: 3,975 (16.78%)
  - Others: 1,692 (7.14%)

3.1.3 FairFace Dataset (Validation Phase)

Dataset Specifications:

- Total Images Used: 50,000
- Demographic Categories: 7 race categories
- Task: Binary gender classification with race-based fairness analysis
- Nature: Balanced and diverse facial image dataset

3.2 Experimental Setup and Hyperparameters

3.2.1 Model Architecture

A Convolutional Neural Network (CNN) was employed for all experiments with the following structure:

- Input Layer: Preprocessed images (normalized to $[0, 1]$)
- Convolutional Layers: Multiple convolution layers with ReLU activation
- Pooling Layers: Max pooling for dimensionality reduction
- Fully Connected Layers: Dense layers for classification
- Output Layer: Binary classification (2 units with softmax activation)

3.2.2 Federated Learning Configuration

Parameter	**Value**
Number of Clients	10
Communication Rounds	8-10
Local Epochs	5
Batch Size	32
Aggregation Method	FedAvg
Device	CUDA (GPU)

Chapter 4

Phase 1: Foundational Federated Learning Understanding

4.1 Objectives and Motivation

Phase 1 focused on establishing foundational understanding of Federated Learning mechanics, particularly:

1. Understanding how FL aggregation works compared to centralized learning
2. Observing the impact of Non-IID data distribution on convergence
3. Understanding FL vulnerability to poisoning attacks
4. Demonstrating how even a single malicious client can severely degrade model performance

4.2 Experimental Design

4.2.1 Experiment 1: Baseline FL with IID Data Distribution

Setup:

- Data randomly distributed across 5 clients (IID assumption)
- Standard FL training for 10 rounds
- No attacks or fairness interventions

Results:

- Global model achieved good convergence
- All clients contributed balanced gradient updates
- Convergence was relatively fast due to data homogeneity

Key Insight: Under IID conditions, FL behaves similarly to distributed SGD and converges relatively smoothly.

4.2.2 Experiment 2: Non-IID Data Distribution Impact

Setup:

- Data deliberately distributed non-uniformly across 5 clients
- Each client assigned only 2 out of 10 digit classes
- Standard FL training without attacks

Results:

- Convergence significantly slower
- Most clients converged to poor local optima
- Class 1 Accuracy: Improved from 0% to 98% on designated client but unstable globally
- Global Model Accuracy: $\approx 73\%$ (compared to typical $> 95\%$ in centralized MNIST)

Key Insight: Non-IID data distribution is problematic for FL convergence and accuracy.

4.2.3 Experiment 3: Label-Flipping Attack on Non-IID Data

Attack Configuration:

- Single malicious client: Only source of Class 1 data
- Attack method: Flip all Class 1 labels to Class 7
- Other 4 clients train normally
- Duration: 10 rounds of FL training

Results:

- Class 1 Accuracy: Dropped from 98% to 0%
- Global model learned incorrect mapping: Class 1 features → Class 7 output
- Accuracy on Class 7: Significantly degraded due to feature confusion

Key Insights:

1. Even a single malicious client can catastrophically corrupt learning
2. Non-IID data distribution amplifies the impact of poisoning attacks
3. FL lacks inherent defense against poisoning without explicit robustness mechanisms
4. In real-world scenarios, such attacks could selectively harm underrepresented groups

4.3 Transition to Fairness-Focused Research

The Phase 1 experiments demonstrated two critical insights:

1. Non-IID data distribution is inevitable and problematic in real FL settings
2. FL systems are fragile to adversarial manipulation, particularly for underrepresented data portions

These observations motivated the hypothesis that FL systems might inherently exhibit fairness issues for demographic minority groups, even without explicit attacks. This led to Phase 2, focusing specifically on fairness analysis in the UTKFace dataset.

Chapter 5

Phase 2: Fairness Issues Identification

5.1 Objectives

Phase 2 aimed to:

1. Establish a baseline FL model on realistic facial data
2. Quantify fairness disparities (race-wise accuracy gaps)
3. Understand the relationship between data imbalance and fairness issues
4. Investigate how adversarial attacks amplify fairness problems

5.2 Baseline FL Results (Clean Setup)

5.2.1 Overall Performance

- Initial Accuracy: $\approx 76.8\%$
- Final Accuracy: $\approx 87.3\%$
- Trend: Steady improvement over 10 rounds

5.2.2 Race-Wise Accuracy Analysis

Detailed Results:

Fairness Gap Calculation:

$$\text{Fairness Gap} = 0.915 - 0.829 = 0.0852 \tag{5.1}$$

Race	Accuracy	Dataset %
White	0.885	42.49%
Black	0.913	19.10%
Asian	0.829	14.49%
Indian	0.915	16.78%
Others	0.863	7.14%

Critical Observation: Despite achieving strong overall accuracy (87.3%), the model exhibits significant race-wise disparity. The Asian demographic group suffers most, with accuracy 8.6 percentage points lower than the highest-performing group.

5.3 Adversarial Attack Scenario

5.3.1 Attack Configuration

- Malicious Clients: 30% (3 out of 10 clients)
- Attack Method: Random label flipping
- Duration: All 10 federated rounds

5.3.2 Results Under Attack

Overall Accuracy Behavior:

- Round 1: Accuracy drops to 0.542 (severe initial degradation)
- Subsequent Rounds: Accuracy fluctuates between 0.60-0.75
- Final Accuracy: ≈ 0.835

Race-Wise Impact:

Race	Clean	Attacked	Loss
White	0.885	0.716	0.169
Black	0.913	0.769	0.144
Asian	0.829	0.691	0.138
Indian	0.915	0.768	0.147
Others	0.879	0.723	0.156

Key Finding: Adversarial attacks disproportionately harm minority racial groups and those with weaker baseline performance.

Chapter 6

Phase 3: Fairness Improvement Methods

6.1 Overview of Fairness Intervention Strategies

Five distinct fairness improvement methods were developed, tested, and evaluated:

1. Static Weighted Cross-Entropy (Global)
2. Fairness Regularization
3. Adversarial Fairness Learning
4. Dynamic Client-Adaptive Fairness Weighting (DCA-FW)
5. Attack-Based Fairness Approach

6.2 Method 1: Static Weighted Cross-Entropy

6.2.1 Concept and Implementation

Mathematical Formulation:

Let w_r denote the weight assigned to racial group r. The weighted cross-entropy loss is:

$$\mathcal{L}_{\text{weighted}} = \sum_r w_r \cdot \mathcal{L}_{\text{CE},r} \tag{6.1}$$

where $\mathcal{L}_{\text{CE},r}$ is the cross-entropy loss for group r.

Weight Calculation:

$$w_r = \frac{1}{\text{baseline_accuracy}_r + \epsilon} \tag{6.2}$$

6.2.2 Results

- Fairness Gap: No significant improvement
- Overall Accuracy: Reduced by 2-3%
- Fairness Metrics (SPD/EOD): Minimal improvement

6.2.3 Analysis of Failure

Why Static Weighting Failed:

1. Weak Adaptation: Weights are static and based on dataset statistics
2. No Real Signal: Assigning weight does not ensure better feature learning
3. Dilution in FL: Weights are averaged during FedAvg aggregation
4. Accuracy-Fairness Trade-off: Forces accuracy degradation

6.3 Method 2: Fairness Regularization

6.3.1 Concept and Implementation

Mathematical Formulation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{fairness}} \tag{6.3}$$

where the fairness penalty is:

$$\mathcal{L}_{\text{fairness}} = \max_r(\mathcal{L}_r) - \min_r(\mathcal{L}_r) \tag{6.4}$$

6.3.2 Results

- Fairness Gap: Worsened to 0.1013
- Overall Accuracy: Dropped by 2-3%
- Fairness Metrics: Minimal improvement

6.3.3 Analysis of Failure

Why Fairness Regularization Failed:

1. Loss vs. Accuracy Mismatch: Minimizing loss disparity does not guarantee accuracy improvement
2. Dilution in FL Convergence: Regularization signal is diluted during aggregation
3. Hyperparameter Sensitivity: Finding optimal λ is difficult
4. Conflict with Aggregation: Different clients have conflicting objectives

6.4 Method 3: Adversarial Fairness Learning

6.4.1 Concept and Implementation

Two-Network Architecture:
Two networks are trained simultaneously in opposition:

Predictor Network P: Predicts gender, minimizes race predictability

Adversary Network A: Predicts race from learned features, forces race-invariance

Formal Objective:

$$\min_{P} \max_{A} \left[\mathcal{L}_{\text{CE}}(Y, P(X)) - \lambda \cdot \mathcal{L}_{\text{Adv}}(Z, A(h(X)))\right] \tag{6.5}$$

6.4.2 Results

- Fairness Gap: Reduced to 0.0736 (improvement from 0.0852)
- Overall Accuracy: Decreased to $\approx 85\%$
- Fairness Metrics: Moderate improvement

6.4.3 Analysis of Performance

Partial Success:

1. Theoretically Sound: Well-grounded in fairness literature
2. Race-Invariant Features: Successfully reduces demographic bias
3. Slight Accuracy Trade-off: Small cost in overall accuracy

Limitations in FL:

1. Increased Complexity: Larger model, higher communication overhead
2. Training Instability: Minimax optimization is notoriously unstable
3. Accuracy-Fairness Trade-off: Does not match DCA-FW's performance

6.5 Method 4: Dynamic Client-Adaptive Fairness Weighting (DCA-FW)

6.5.1 Concept and Mechanism

Core Idea:

At each federated round, designated "fairness clients" perform:

1. Compute local race-wise accuracy after local training
2. Identify underperforming racial groups
3. Adjust loss weights inversely to local performance
4. Train with updated weights next round

Mathematical Formulation:

Let $a_r^k(t)$ denote the local race-wise accuracy for group r at client k in round t:

$$w_r^k(t+1) = \frac{1}{a_r^k(t) + \epsilon} \tag{6.6}$$

In the next round, the weighted loss at client k becomes:

$$\mathcal{L}_k(t+1) = \sum_r w_r^k(t+1) \cdot \mathcal{L}_{\mathrm{CE},r}^k(t+1) \tag{6.7}$$

6.5.2 Key Innovations

1. **Client-Level Adaptation:** Weights computed locally, preventing dilution
2. **Dynamic Updates:** Weights change every round based on actual performance
3. **Per-Client Customization:** Different clients use different weights
4. **Reduced Overhead:** Only 30% of clients are fairness clients

6.5.3 Results

UTKFace Results:

- Fairness Gap: 0.0852 → 0.0507 (40% reduction)
- Overall Accuracy: 0.873 (maintained)
- Fairness Metrics: Significant improvement

Race-wise Accuracy:

Race	Baseline	DCA-FW
White	0.885	0.879
Black	0.913	0.911
Asian	0.829	0.860
Indian	0.915	0.906
Others	0.863	0.869

6.5.4 Why DCA-FW Succeeds

Success Factors:

1. **Aligned with FL Architecture:** Operates at client level, avoids aggregation dilution
2. **Performance-Based Adaptivity:** Based on actual model performance, not static data
3. **Each Client as Agent:** Harnesses natural diversity of FL
4. **Direct Targeting:** Increases loss weights for underperforming groups
5. **No Accuracy Trade-off:** Works within existing training objective
6. **Robust to Non-IID Data:** Benefits from Non-IID data independence

6.5.5 Fairness Client Selection

Fairness clients were selected randomly at the beginning (30% of total clients). Alternative selection strategies include:

- Capacity-Based: Based on computational resources
- Adaptive: Dynamically change membership based on performance
- Responsibility-Based: Based on demographic service
- Voluntary: Client opt-in based on organizational policy

6.5.6 Weight Computation Details

Local Accuracy:

$$a_r^k = \frac{\text{correct_predictions}_r}{\text{total_samples}_r} \tag{6.8}$$

Weight Update:

$$w_r^k = \frac{1}{a_r^k + \epsilon}, \quad \epsilon = 0.01 \tag{6.9}$$

6.5.7 Global Model Aggregation

$$\mathbf{w}^{(t+1)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathbf{w}_k^{(t+1)} \tag{6.10}$$

6.6 Method 5: Attack-Based Fairness Approach

6.6.1 Concept and Motivation

Hypothesis: Adversarial attack behavior could reduce fairness gaps by degrading high-performing demographic groups.

6.6.2 Results

- Fairness Gap: Worsened to 0.1064
- Overall Accuracy: Relatively maintained
- Behavior: Gap increased rather than decreased

6.6.3 Analysis of Failure

Why Attack-Based Fairness Failed:

1. Attacks Are Not Precision Instruments: Affect all groups indiscriminately
2. FL Averaging Dilution: 30% attacks diluted by 70% clean updates
3. Attacks Hurt All Groups: Corrupt global model for all demographics
4. Counterintuitive Result: High-performers more resilient, gap increased
5. Theoretical Mismatch: Fairness is constructive, not destructive

6.7 Comparative Summary of All Methods

Method	**Fairness Gap**	**Overall Acc**	**Outcome**
Baseline	0.0852	0.873	Biased
Static Weighting	0.085±	0.845-0.855	Ineffective
Fairness Regularization	0.1013	0.850-0.860	Worsened
Adversarial Learning	0.0736	0.850-0.860	Partial Success
Attack-Based	0.1064	0.873-0.884	Failed
DCA-FW	0.0507	0.873	**Optimal**

Chapter 7

Phase 4: Validation on FairFace Dataset

7.1 Motivation for Multi-Dataset Validation

Single-dataset evaluation risks overfitting conclusions to that dataset's specific characteristics. FairFace provides:

- Different Race Categories: 7 racial categories vs. 5 in UTKFace
- Different Data Source: Independently collected facial images
- Better Demographic Balance: More balanced representation
- Larger Dataset: 50,000 images (larger than UTKFace's 23,695)

7.2 FairFace Baseline FL Results

Configuration:

- Images Selected: 50,000 (from 100,000+ available)
- Train/Test Split: 80/20
- Clients: 10, Rounds: 10

Results:

- Overall Accuracy: 0.7723 (77.23%)
- Fairness Gap: 0.0992

Race-wise Accuracy:

Race	Accuracy	Rank
White	0.7645	3
Black	0.7226	7 (Lowest)
East Asian	0.7373	5
Indian	0.8056	2
Latino_Hispanic	0.8145	3
Middle Eastern	0.8218	1 (Highest)
Southeast Asian	0.7469	4

7.3 FairFace with DCA-FW - Multiple Runs

7.3.1 Run 1: Fairness Clients {0, 1, 4, 9} (40%)

Results:

- Average Accuracy: 0.7790 (77.90%)
- Fairness Gap: 0.0880
- Gap Reduction: 0.0992 → 0.0880 (11.2% reduction)
- Accuracy Change: +0.67%

7.3.2 Run 2: Fairness Clients {0, 1, 4, 6, 9} (50%)

Results:

- Average Accuracy: 0.7695 (76.95%)
- Fairness Gap: 0.0858
- Gap Reduction: 0.0992 → 0.0858 (13.5% reduction)
- Accuracy Change: -0.28%

7.3.3 Run 3: Fairness Clients {0, 1, 4, 5, 6, 9} (60%)

Results:

- Fairness Gap: 0.0812
- Gap Reduction: 0.0992 → 0.0812 (18.1% reduction)
- Outcome: Best fairness gap achieved

7.4 Analysis of Multi-Run Consistency

Key Observations:

1. **Consistent Improvement:** All runs reduce fairness gap versus baseline
2. **Dose-Response Relationship:** More fairness clients → greater fairness improvement
3. **Minimal Accuracy Trade-off:** All runs maintain accuracy within 0.95% range
4. **Generalization Confirmed:** DCA-FW effectiveness validates across different datasets

Chapter 8

Conclusions and Future Work

8.1 Key Findings

8.1.1 Federated Learning Inherently Exhibits Fairness Disparities

Even at baseline without adversarial intervention, standard FL models exhibit significant race-wise accuracy gaps. This suggests that FL, by virtue of its Non-IID distributed nature and client-level processing, naturally amplifies demographic biases.

8.1.2 Traditional Fairness Methods Fail in FL Settings

Static weighting, global regularization, and adversarial learning fail in FL due to:

- Aggregation dilution
- Conflict between local and global objectives
- Hyperparameter sensitivity
- Accuracy-versus-fairness trade-offs

8.1.3 Client-Level Adaptive Mechanisms are Essential

DCA-FW succeeds because it operates at individual client level, the fundamental FL unit, rather than attempting global constraints.

8.1.4 Fairness Improvement Without Accuracy Sacrifice

DCA-FW achieves simultaneous fairness improvement and accuracy maintenance, representing an optimal solution.

8.1.5 Method Generalizes Across Datasets

Validation on FairFace confirms DCA-FW's applicability to different facial datasets and demographic distributions.

8.2 Implications for Federated Learning Systems

8.2.1 Design Implications

1. Fairness must be integrated into FL system architecture from inception
2. Client-level instrumentation is necessary for demographic tracking
3. Transparency and fairness client designation should be clearly communicated

8.2.2 Practical Implementation Considerations

1. **Metadata Requirements:** Clients must track demographic attributes
2. **Computational Overhead:** Minimal; local accuracy evaluation is standard
3. **Communication Efficiency:** No increase in model size or communication

8.3 Limitations

1. Limited to binary gender classification task
2. Race-based fairness focus; other attributes unexplored
3. Fairness client ratio requires empirical determination
4. Limited to 8-10 communication rounds
5. Basic label-flipping attacks tested; sophisticated poisoning unexplored

8.4 Future Research Directions

8.4.1 Technical Extensions

1. **Multi-Class Tasks:** Extension to multi-class classification with multiple protected attributes
2. **Adaptive Selection:** Dynamically adjust fairness client membership
3. **Alternative Weight Functions:** Beyond $1/(a_r + \epsilon)$ formulation
4. **Theoretical Analysis:** Convergence guarantees for DCA-FW

8.4.2 Application Domains

1. **Non-Classification Tasks:** Regression, ranking, recommendation systems
2. **Multiple Protected Attributes:** Intersectional fairness analysis
3. **Heterogeneous Networks:** Variable bandwidth and latency conditions

8.5 Final Summary

This project successfully identified inherent fairness disparities in Federated Learning systems, explored five fairness improvement methods, and developed Dynamic Client-Adaptive Fairness Weighting (DCA-FW), a robust solution that improves fairness by 40% on UTKFace and 18% on FairFace without sacrificing accuracy. The research demonstrates that client-level adaptive mechanisms are key to fairness in decentralized learning systems. Findings are validated across multiple datasets and serve as foundation for future work in federated fairness.

Appendix A

DCA-FW Algorithm Pseudocode

Algorithm 1 Dynamic Client-Adaptive Fairness Weighting

```
for each federated round t do
    Server broadcasts global model w^t to all clients
    for each client k do
        Download global model w^t
        Perform local training on local dataset D_k for E epochs
        if k is a fairness client then
            Evaluate current local model on D_k
            Compute local race-wise accuracies a_r^k
            Compute weight adjustments: w_r^k(t+1) = 1/(a_r^k + ε)
            Update loss function: L_k ← weighted_Loss(w_r^k(t+1))
        end if
        Send updated model w_k^(t+1) to server
    end for
    Server aggregates: w^(t+1) = Average(w_k^(t+1))
    Evaluate global model on global test set
end for
```

Appendix B

Results Summary Tables

B.1 Complete Results Comparison

Dataset/Method	Fairness Gap	Accuracy	Status
UTKFace			
Baseline	0.0852	0.8730	Significant Bias
Static Weighting	0.085 ± 0.01	0.845–0.855	Ineffective
Fairness Regularization	0.1013	0.850–0.860	Worsened
Adversarial Learning	0.0736	0.850–0.860	Partial Solution
Attack-Based	0.1064	0.873–0.884	Failed
DCA-FW	0.0507	0.8730	**Optimal**
FairFace (Multiple Runs)			
Baseline	0.0992	0.7723	Significant Bias
DCA-FW (30%)	0.0880	0.7790	Strong Improvement
DCA-FW (40%)	0.0858	0.7695	Better Fairness
DCA-FW (50%)	0.0812	*Not logged*	Best Gap